

(I1 - Importance of tests.)

Stability Test: The stability test will be addressing the Stability/Structure design requirement found in Element C. This test evaluates whether or not the desk is able to function like a normal desk. It looks at whether the desk is able to hold weight, is it stable when in use, can it store documents and other necessary materials, and more. This test is important because it allows us to see the areas in which our desk needs improvement and which areas we need to be extra cautious with when building our actual model.

Aesthetics Test: The aesthetics test will be addressing a few of our high priority design requirements detailed in Element C. These include the resin art on the desk, the power strip under the monitor shelf, and the monitor shelf. These aspects of the project are incredibly important since one of the main goals of this project is to improve the work ethic of Ms. A.V by providing a comfortable and pleasing work environment for her. This test will evaluate how the different stakeholders feel about the aesthetics of the desk and what aspects are unpleasant to them so that we can work to improve these aspects when creating the actual model.

(I2 - Comprehension of testing procedure)

Stability Test:

Independent Variable(s): Weight that is put on the desk

Dependent Variable(s): How well the desk holds up

Procedure:

- Gather weights to put on desk
- Place prototype desk on a flat, even surface
- Place weights on top of the desk and analyze the structure of the desk. Analyze how the tabletop is holding up, the stability of the legs, etc.
- Document these observations down to present to Mentor Stakeholders
- Ask Mentor Stakeholders for advice on how we should build these mechanics and functions on our actual model. Document their thoughts

Aesthetics Test:

Independent Variable(s): The stakeholders reviewing the looks of the desk

Dependent Variables(s): Whether or not the desk is aesthetically pleasing

Procedure:

- Make a google form / survey that allows for Stakeholders to express their opinions on the looks of the desk
- Analyze the responses from the Stakeholder and document them
- Summarize the feedback to present to Ms. A.V (Primary Stakeholder). Ask for her opinion on how she feels about the advice and if she approves, apply these changes to the actual model.

Durability Test:

Independent Variable(s): Expose prototype to physical tests including dropping from a height, shaking, etc.

Dependent Variable(s): How well the desk holds up after physical tests

Procedure:

- First we will do the shaking test.
- Grab desks from different areas including table legs, monitor shelf, cabinet etc. Shake the desk for a minimum of 5 sec.
- Observe the condition of the desk. Note down areas of the desk that are unstable or have been damaged.
- Then proceed to the dropping test. Drop the desk from a height of 2 ft. multiple times.
- After each drop, observe the condition of the desk. Note how the desk has responded to impact, any damage it sustained, and aspects of the desk that seem unstable.
- Summarize the data from the drop and shake test to present to Senior Stakeholders.
- Ask for their opinion on how, when creating this actual model, we can avoid these issues and create a stable, safe desk for Ms. A.V , our Primary Stakeholder.

(I3 - Detail of the design analysis.)

Stability (In order of the items placed on the desk. No items were removed and all were on top of the desk at the same time)	Aesthetics	Durability
11. Phone - Placed phone on the main part of the desk - Observed no visible change to the structure of the desk	1. C.G - The areas where the cardboard connects are visible, making it look rough. - Should have used more colors when creating the desk to properly represent	21. Shake Test - Grabbed from cabinet - Note that there was initial damage to the legs. There was an issue where the legs were connected with the main desk. The construction paper was

	the Stakeholder's wants.	peeling off, resulting in the legs still being attached to the paper but not necessarily the table - When shake, the desk was fine but the legs had taken slight damage
12. Phone - Placed phone on the monitor shelf - Observed no visible change to the stability of the shelf	2. V.T - Had a concern about how the resin river improperly converts - Didn't create the resin river that will go on the side of the cabinet, making the prototype model plain and doesn't represent how the actual model will look	22. Shake Test - Desk was held by the main surface of the desk - There was no issue with the other aspects of the desk, other than the legs, as mentioned before.
13. Purse - Placed purse on the main part of the desk - Observed no visible changes to the stability of the desk	3. K.Z - Had a concern about the monitor shelf not properly representing the 3D model and sketches. The monitor shelf is supposed to be made completely of resin but our prototype presents it as if it's supposed to be made of wood. - The edges of the desk were also visible, giving it a rough and messy appearance	23. Shake Test - The desk was held by the monitor shelf - There was slight damage to the attachment of the monitor shelf
14. Phone - Placed phone on the monitor shelf - Witnessed slight bend but no significant strain	4. O.B - The brown color of the desk makes it look plain and doesn't convey the materials that we will use for the actual project. - The color of the desk looks matte while the actual	24. Shake Test - Held from main part of the desk - No visible damage

	model will look more glassy due to final touches	
15. Phone - Placed phone on the main part of the desk - Observed no visible strain	5. Mr. P - Edges of the project looked a bit rough	25. Shake Test - Was going to be held from the legs but due to the initial damage, we didn't want to risk damaging the paper completely so we avoided this part of the text - Analyzed the damage legs and saw that the damage came from where it was attached with construction paper
16. Chromebook - Placed chromebook on monitor shelf - Moved phones on the main part of the desk to create an even platform to rest chromebook on - Witnessed no visible strain	6. Mr. B - Had concerns about whether or not or real model would still have the resin waterfall as described to him initially	26. Drop Test - Drop from a height of 5 ft - Legs of the desk completely fell off upon impact to the floor - No other damage to the desk visible
17. Pencil Case - Placed pencil case on monitor shelf - No visible strain	7. Ms. V - The desk wasn't an accurate version of the prototype decision that was said to me as the stakeholder - It makes me believe the resin will be white, but I wish to have clear resin with white inclusions - Also, the knobs on the pull-out drawers would be nice if there were like clear crystal or just clear plastic or gold to make it look nice. - Overall, it's love for the desk and the vision there is	27. Drop Test - Dropped from a height of 2 ft - Did not reattach the legs to the desk - No other damage to the desk visible

	for the desk.	
18. Headphones - Place on the top of the desk - The desk did not take nay damage front he wirght of the headphones and was able to withstand it for 10 whole mintues.l	8. A.B - Had concerns about how the color of the prototype for the resin river would be reflected on the actual model	28. Drop Test - Dropped from a height of 3 ft - Slight damage to the flower on top of the monitor shelf
19. Chromebook x3 - Placed remaining chromebooks on top of the first one - Witnesses the monitor shelf bend significantly but did not break	9. A.G - Liked the prototype model but had questions on how we would install the drawers on the	29. Drop Test - Dropped from 2 ft - No damage
20. Wallet - Placed wallet on top of monitor shelf - Witnessed no visible strain on monitor shelf after being exposed to the model	10. S.L - Liked the design of the desk - Asked about what would the true color of the resin river be	30. Drop Test - Dropped from a height of 3 ft - Slight damage to the corner of the desk visible

(14 - Summary of data implications.)

Stability Test:

- There were a few concerns about the stability of the desk
- The height of the legs and the cabinet were unequal, resulting in the desk being uneven
- This issue can be fixed by using table glides, as recommended by Olivia. This can create an adjustable desk that allows for the height to be fixed. This also prevents the desk from creating scratch marks on the floor

Aesthetics Test:

- There were a couple of concerns about the prototype desk
- The prototype desk did not have the resin design on the legs of the desk so it did not show the full idea of our desk design.

Resin River:

- Concerns about it not being completely represented on the prototype
- Stakeholder, after seeing the model, had a couple of comments about it
- The Stakeholder requested that the resin river be clear, with white inclusions
- Additionally, Ms. A.V requested a variety of petals instead of just roses
- We need to communicate these changes and concerns with Mr. R.N, who is our mentor when we're working with resin

Wood:

- Color of the prototype doesn't properly convey the actual color and texture of the wood
- Actual wood during the project will have a more smooth and sleek appearance due to the finishings that will be used